

SATURNA CAPMON, BC, Canada

WMO: 719140

Lat: 48.7753N

Lon: 123.1281W

Elev: 584

StdP: 14.39

Time Zone: -8.00 (NAP)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	23.8	26.7	7.2	8.1	28.6	12.8	10.8	30.0	11.7	39.8	10.0	38.7	4.1	60	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.1	80.0	62.2	76.7	60.9	73.6	60.0	64.2	74.4	62.5	72.9	61.2	70.9	2.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.3	79.9	68.0	58.5	74.7	65.4	57.2	71.3	63.9	29.6	74.8	28.4	73.2	27.4	71.0	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.1	5.6	DB	20.2	86.2	3.4	4.1	17.8	89.2	15.8	91.5	14.0	93.8	11.5	96.8	(4)
(5)			WB	17.1	65.9	3.8	3.4	14.4	68.3	12.2	70.3	10.0	72.2	7.3	74.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	39.3	39.6	42.9	47.1	53.9	57.7	63.3	63.7	58.7	50.1	43.2	37.9	(6)	
	DBStd	10.48	5.06	5.63	4.76	4.71	5.55	5.53	5.45	5.12	5.37	4.17	5.53	5.54	(7)	
	HDD50	1616	331	290	224	112	20	1	0	0	1	49	212	376	(8)	
	HDD65	5656	795	710	684	537	348	233	100	89	202	462	655	841	(9)	
	CDD50	1563	0	0	5	24	140	234	412	426	263	52	7	0	(10)	
	CDD65	129	0	0	0	0	3	16	46	50	14	0	0	0	(11)	
	CDH74	619	0	0	0	2	12	84	260	211	50	0	0	0	(12)	
	CDH80	98	0	0	0	0	0	12	57	26	3	0	0	0	(13)	
(14) Wind	WSAvg	2.2	2.3	2.2	2.2	2.0	2.1	2.2	2.4	2.1	2.0	2.0	2.5	2.4	(14)	
(15) Precipitation	PrecAvg	33.0	4.8	2.9	3.0	2.0	1.7	1.4	0.9	1.1	1.8	3.7	4.9	4.8	(15)	
	PrecMax	40.8	7.6	6.1	5.9	4.6	3.0	2.7	2.0	4.2	5.4	10.2	8.5	8.1	(16)	
	PrecMin	23.1	2.4	0.5	0.9	0.1	0.0	0.1	0.0	0.1	0.2	0.4	1.8	1.8	(17)	
	PrecStd	4.6	1.4	1.2	1.2	1.0	0.8	0.7	0.5	0.9	1.3	2.0	1.9	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	52.1	59.8	67.9	76.0	81.9	86.2	83.4	79.5	64.4	57.1	52.6	(19)	
		MCWB	49.3	47.4	48.3	52.9	57.7	62.4	65.8	62.0	60.8	55.7	53.3	50.6	(20)	
	2%	DB	49.6	50.1	54.9	61.1	70.8	76.0	80.3	79.2	74.5	60.8	54.4	49.2	(21)	
		MCWB	47.7	46.1	47.0	49.8	56.4	59.7	63.4	61.5	59.3	53.4	52.3	47.3	(22)	
	5%	DB	47.8	48.3	51.9	57.3	67.3	71.0	76.4	76.0	70.7	58.1	52.1	46.8	(23)	
		MCWB	45.9	45.1	45.9	48.5	55.2	58.4	61.9	60.6	58.8	52.7	50.1	44.8	(24)	
	10%	DB	45.9	46.4	49.4	54.0	63.4	66.9	72.7	72.9	67.2	55.8	50.3	44.9	(25)	
		MCWB	44.1	43.7	45.2	46.8	53.6	56.8	60.5	60.2	57.2	51.3	48.3	43.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	49.7	51.2	54.3	60.1	64.8	70.0	65.5	63.4	58.0	54.9	51.6	(27)	
		MCDB	51.1	51.2	56.4	65.0	70.4	75.1	80.6	76.2	73.1	60.7	55.8	52.4	(28)	
	2%	WB	48.5	47.5	48.6	51.6	58.0	61.3	65.4	63.4	61.7	55.9	52.7	47.9	(29)	
		MCDB	49.3	48.9	52.3	58.8	67.7	72.6	76.0	73.8	70.5	58.9	53.8	49.3	(30)	
	5%	WB	46.1	45.7	47.0	49.5	56.2	59.3	62.8	62.2	60.1	53.8	50.4	45.2	(31)	
		MCDB	47.3	47.3	50.4	55.0	64.8	68.6	73.0	72.4	67.7	56.4	51.7	46.5	(32)	
	10%	WB	44.5	44.1	45.7	47.8	54.3	57.4	61.2	60.9	58.7	52.1	48.8	43.3	(33)	
		MCDB	45.7	45.9	48.8	52.6	61.9	65.9	71.0	70.6	64.9	54.8	50.2	44.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	7.4	9.7	11.2	13.3	13.5	14.9	14.1	11.7	8.4	6.9	5.6	(35)	
		MCDBR	7.8	9.4	12.4	15.5	18.3	17.9	18.7	17.6	16.2	11.1	8.0	6.7	(36)	
	5% WB	MCWBR	5.9	6.1	7.2	8.0	8.7	8.0	7.6	7.1	6.7	6.6	6.2	5.9	(37)	
		MCWBR	7.2	8.0	11.0	13.2	16.4	16.1	17.1	16.0	13.8	9.3	7.4	6.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.312	0.307	0.323	0.350	0.370	0.347	0.343	0.349	0.333	0.325	0.313	0.305	(40)	
		taud	2.536	2.575	2.544	2.459	2.419	2.497	2.542	2.524	2.567	2.575	2.545	2.517	(41)	
	Ebn at Noon		242	273	285	286	283	290	289	283	278	262	239	228	(42)	
		Edh at Noon	19	23	27	32	35	33	31	30	26	22	18	17	(43)	
(44) All-Sky Solar Radiation	RadAvg		285	563	867	1328	1713	1860	2048	1731	1219	670	339	228	(44)	
	RadStd		41	79	108	131	140	192	157	121	103	81	56	32	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page